

ouees-202606 topic 04:

- **Internet Protocol (IP) addresses**
- **Routing in details**
- **Network transports**

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On the internet

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Lecture notes and reporting

- <https://github.com/jj1bdx/oueees-202606-public/>
- Check out the README.md file and the issues!
- Keyword at the end of the talk
- URL for submitting the report at the end of the talk

Internet Protocol (IP) addresses

Role of IP addresses

- **Network numbers**
- **Interfaces: connected to the networks**
- **Host IDs in the numbered networks**
- **Global uniqueness**
- **Special addresses (private, broadcast, multicast, loopback, etc.)**

IPv4 addresses: 32 bits

- **192.168.100.20**
- **In hexadecimal notation: 0xC0A86414**
- **4 x 0~255 numbers split with dots**
- **Relatively easy to remember, but already being used up**

IPv4 address with netmask

- **192.168.100.20/24**
- **Network: 192.168.100.0/24**
- **Host: number 20 (0~255) ($32-24=8$)**
- **Host 0 = network itself**
- **Host 255 = broadcast**

Address in another netmask

- **192.168.100.20/28**
- **Network: 192.168.100.16/28**
- **Host: number 4 (0~15) ($32-28=4$)**
- **Host 0 = network itself**
- **Host 15 = broadcast**
- **Different netmask = different address interpretation**

Private addresses (RFC 1918)

No global routing for these address blocks:

- 10.0.0.0/8**
- 172.16.0.0/12 (172.{16~31}.*.*)**
- 192.168.0.0/16 (192.168.*.*)**

Other special addresses (RFC 6890)

- **0.0.0.0/8: "This" network**
- **100.64.0.0/10: Shared address**
- **127.0.0.0/8: Loopback**
- **169.254.0.0/16: Link local**
- **192.0.0.0/24: IANA specific**
- **192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24: Documentation**
- **192.88.99.0/24: 6to4 Relay Anycast**
- **198.18.0.0/15: Benchmarking**
- **240.0.0.0/4: Reserved**
- **255.255.255.255/32: Limited broadcast**

IPv6 addresses: 128 bits

- **2a03:2880:f24e:e0:face:b00c::4420**
- **2a03:2880:f24e:00e0:face:b00c:0000:4420**
- **a www.instagram.com address, as of 17-JUN-2026 0318UTC**
- **:xxxx: = up to 4 hex digits**
- **:: = arbitrary number of 0, appearing only once in an address**
- **Your lookup results may vary**

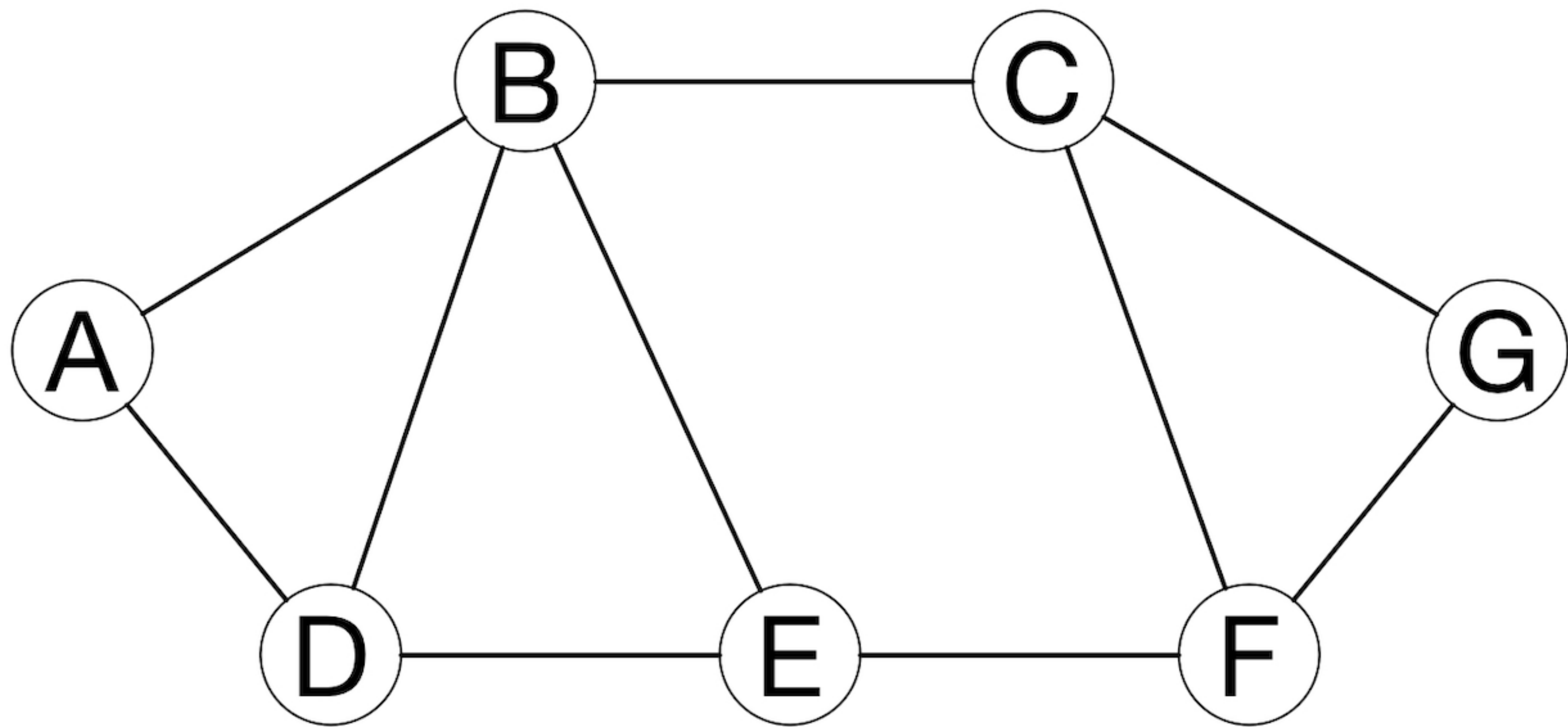
IPv6 addresses with netmask

- **2a03:2880:f24e:e0:face:b00c::4420/64**
- **Network: 2a03:2880:f24e:e0::/64**
- **Host number: 0xfaceb00c00004420**
- **Host number: 64 bits (0: network)**
- **Broadcast -> multicast addresses**
- **ff02::1 = all hosts, ff02::2 = all routers, etc.**

Why IPv4 to IPv6?

- **Because we've used up the 32-bit IPv4 addresses already**
- **No more new address block for IPv4**
- **You need to buy unused blocks from other users**
- **Took ~20 years (1996-2016) for the transition from IPv4 to IPv6**
- **IPv6 has less users and nodes; plausibly less congested**

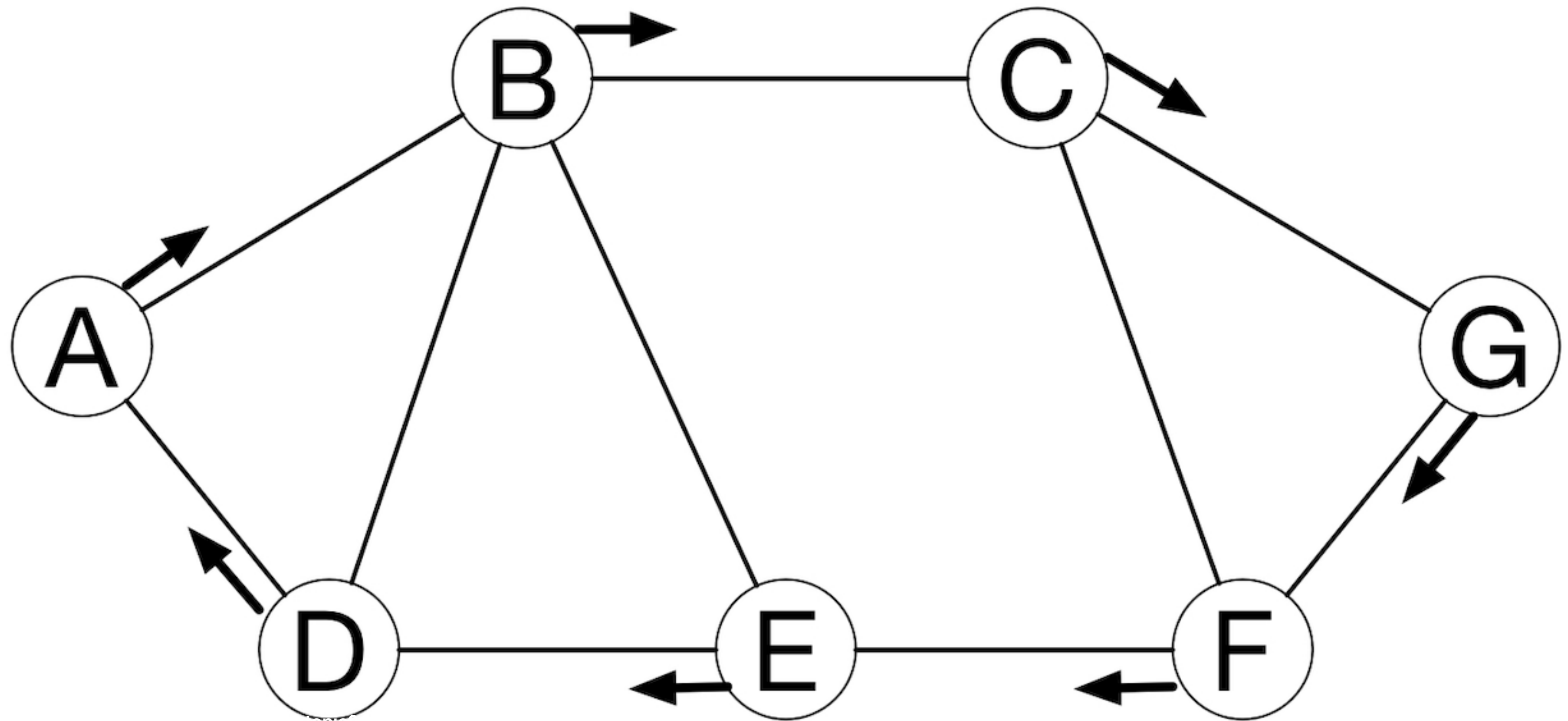
Routing in details



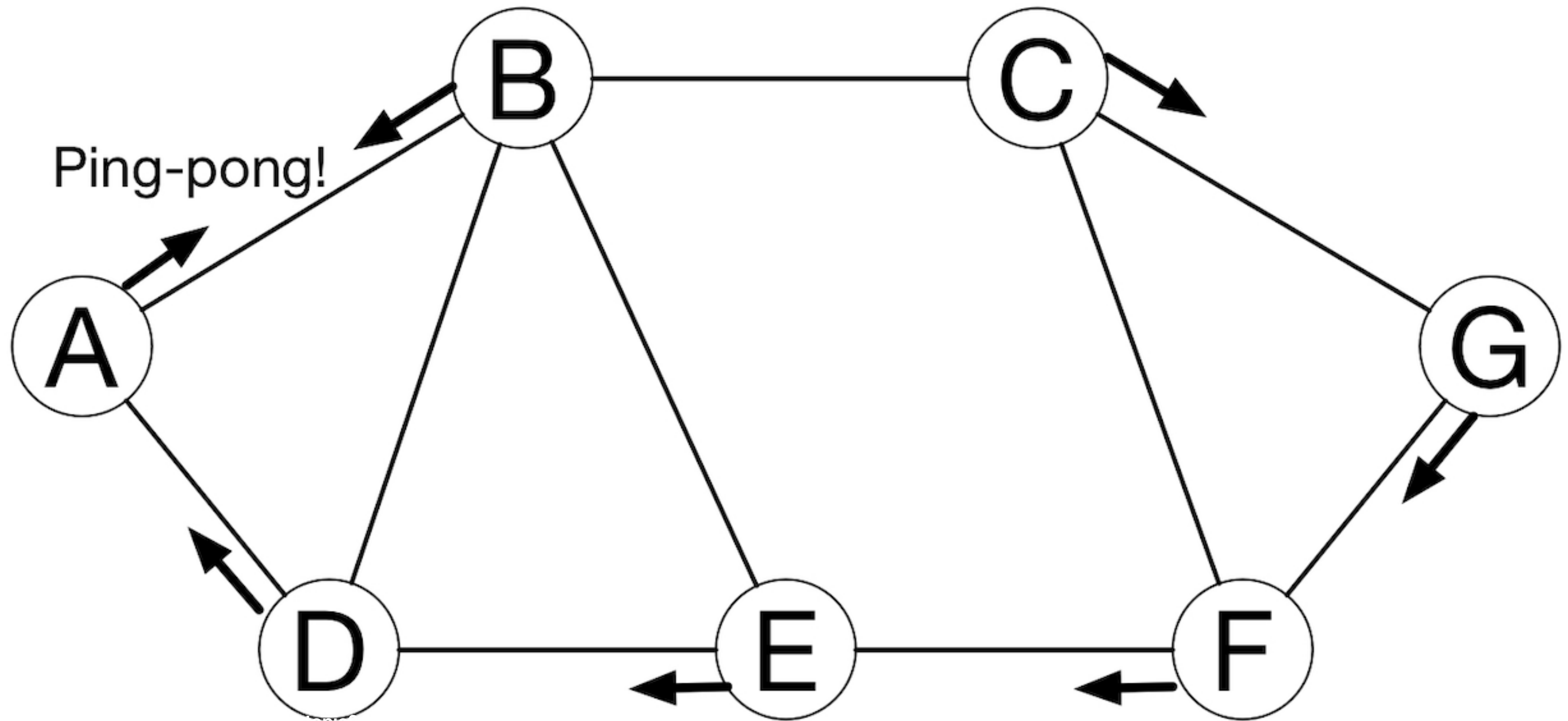
Static routing

- Set the default route for nodes which are not directly reachable
- Works well on simple networks or star networks
- Static routing may cause *ping-pong*

→ Default static route



→ Default static route

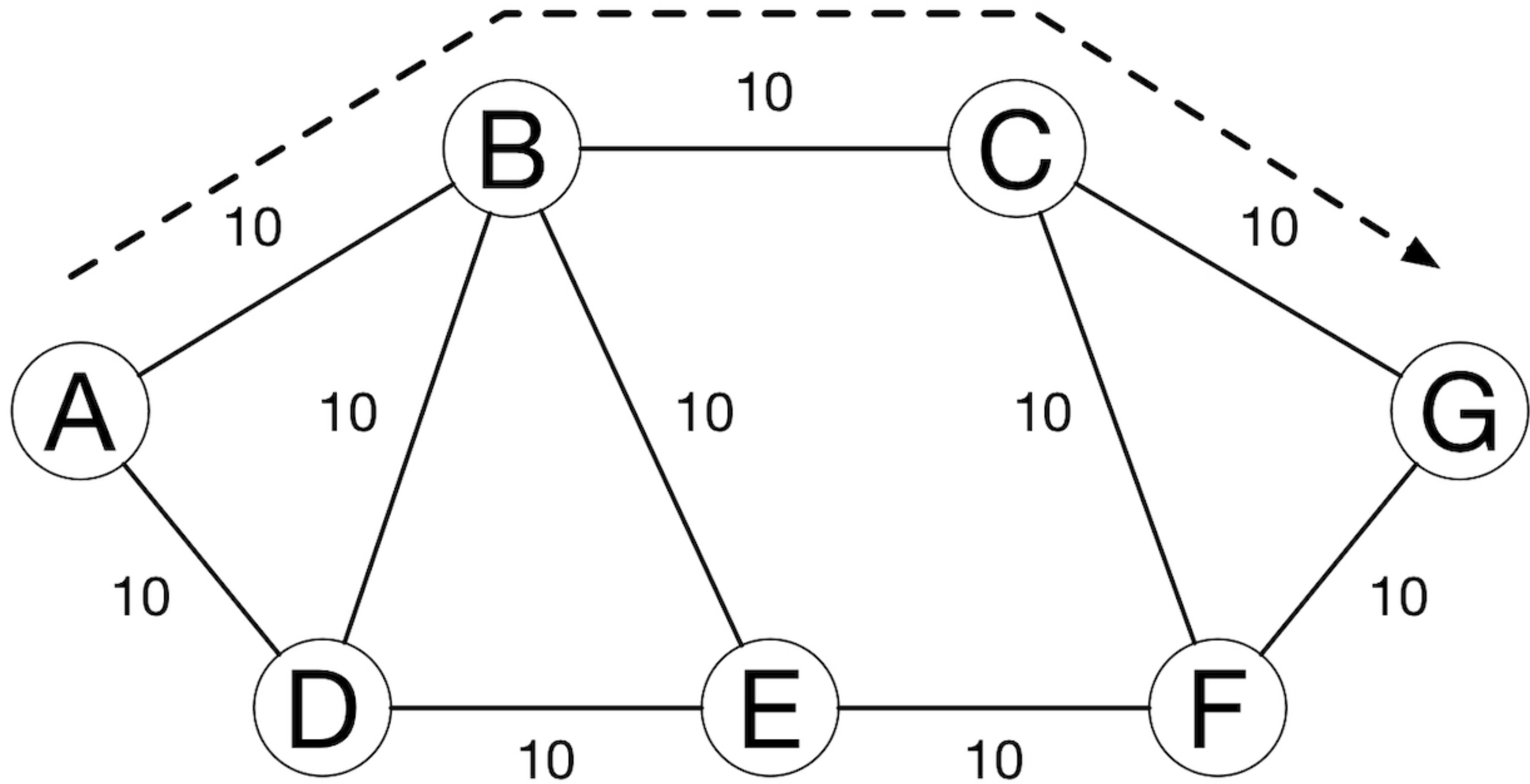


Dynamic routing

- **Hop count: count the hops between nodes**
- **Link cost: determined by the speed and quality**
- **Administrative policies**

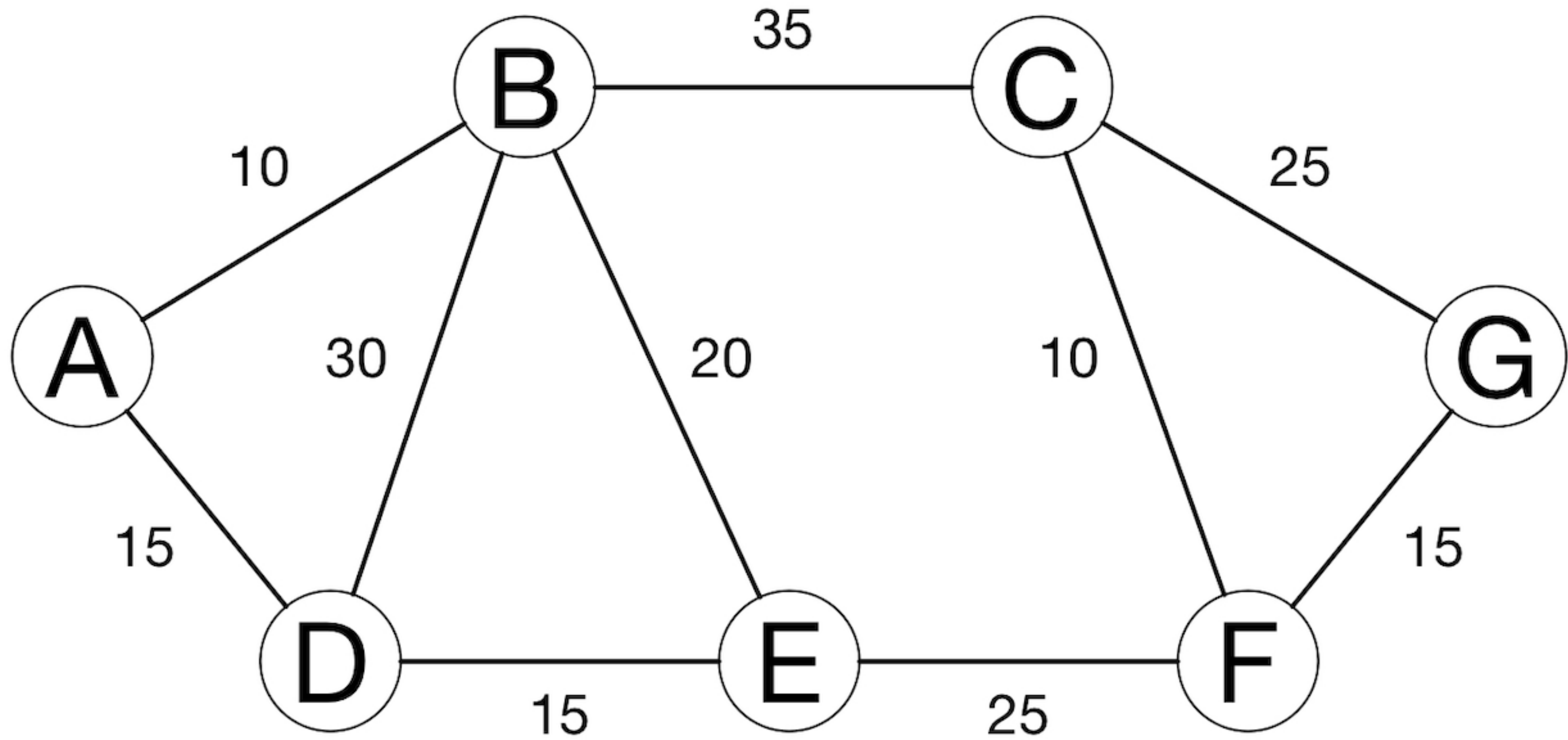
Simple hop counting

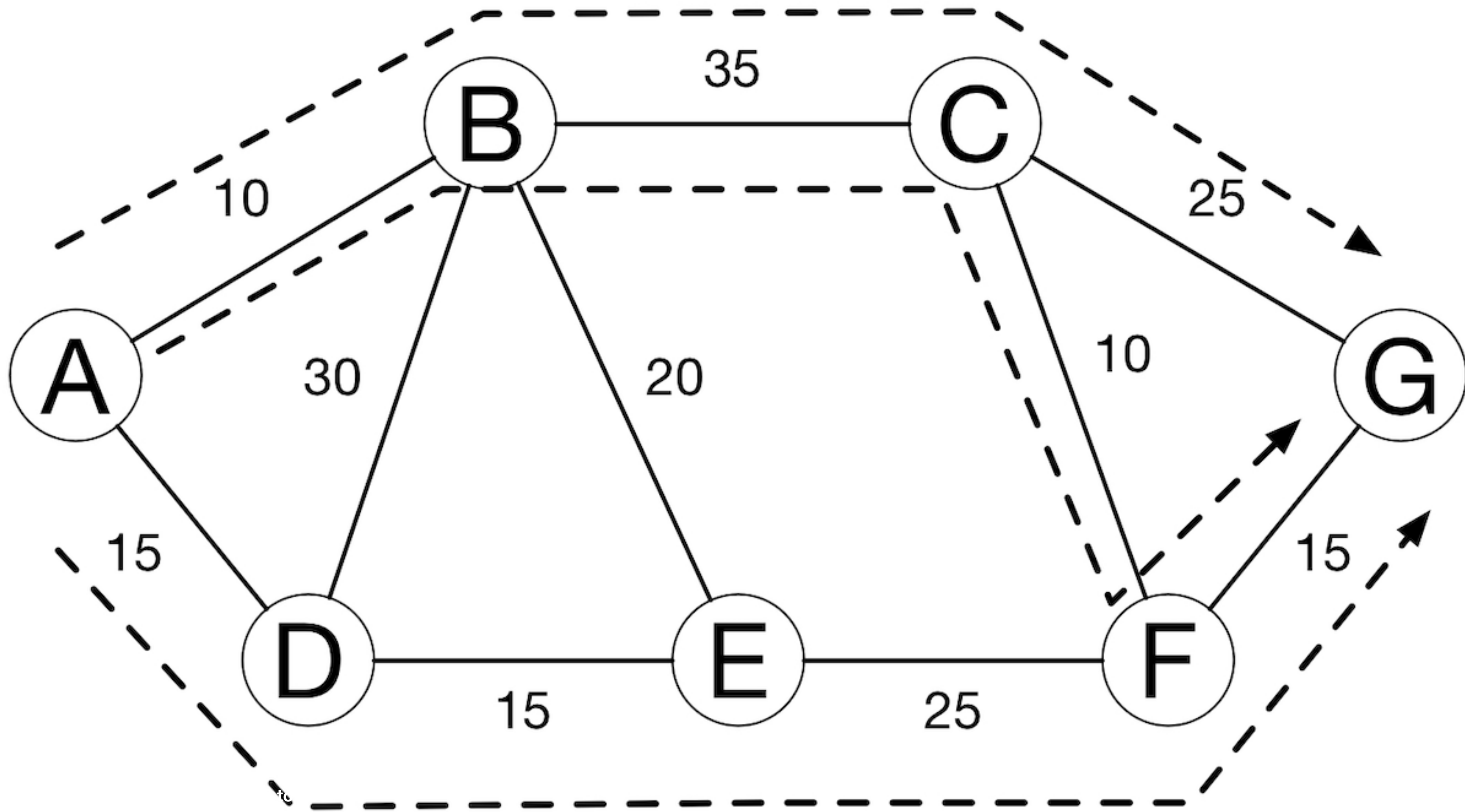
— **Assume every link costs the same with each other**



Evaluating link cost

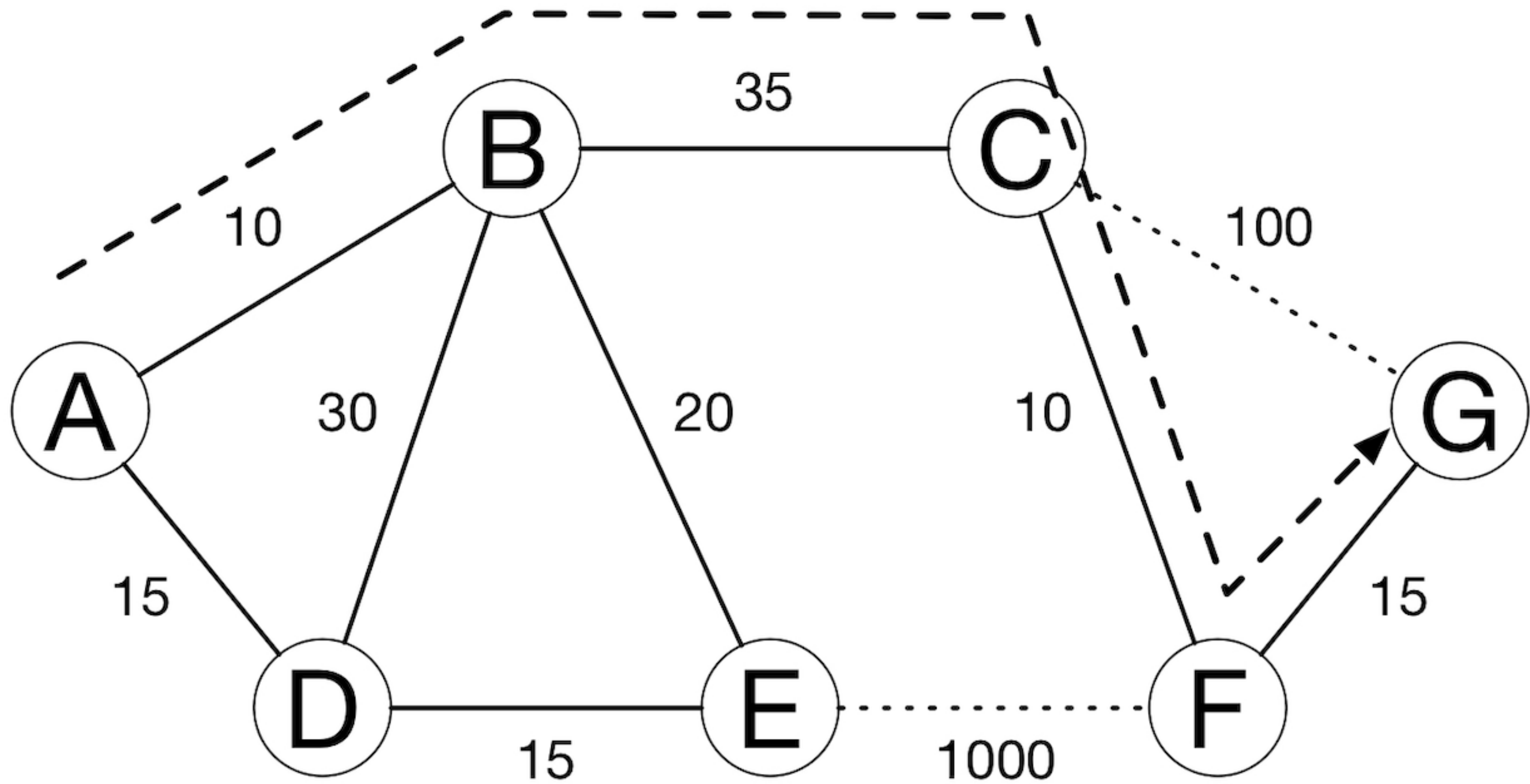
- **What if the cost of each link varies?**
- **If two or more paths have the equal cost, all of the links will be utilized for load balancing**





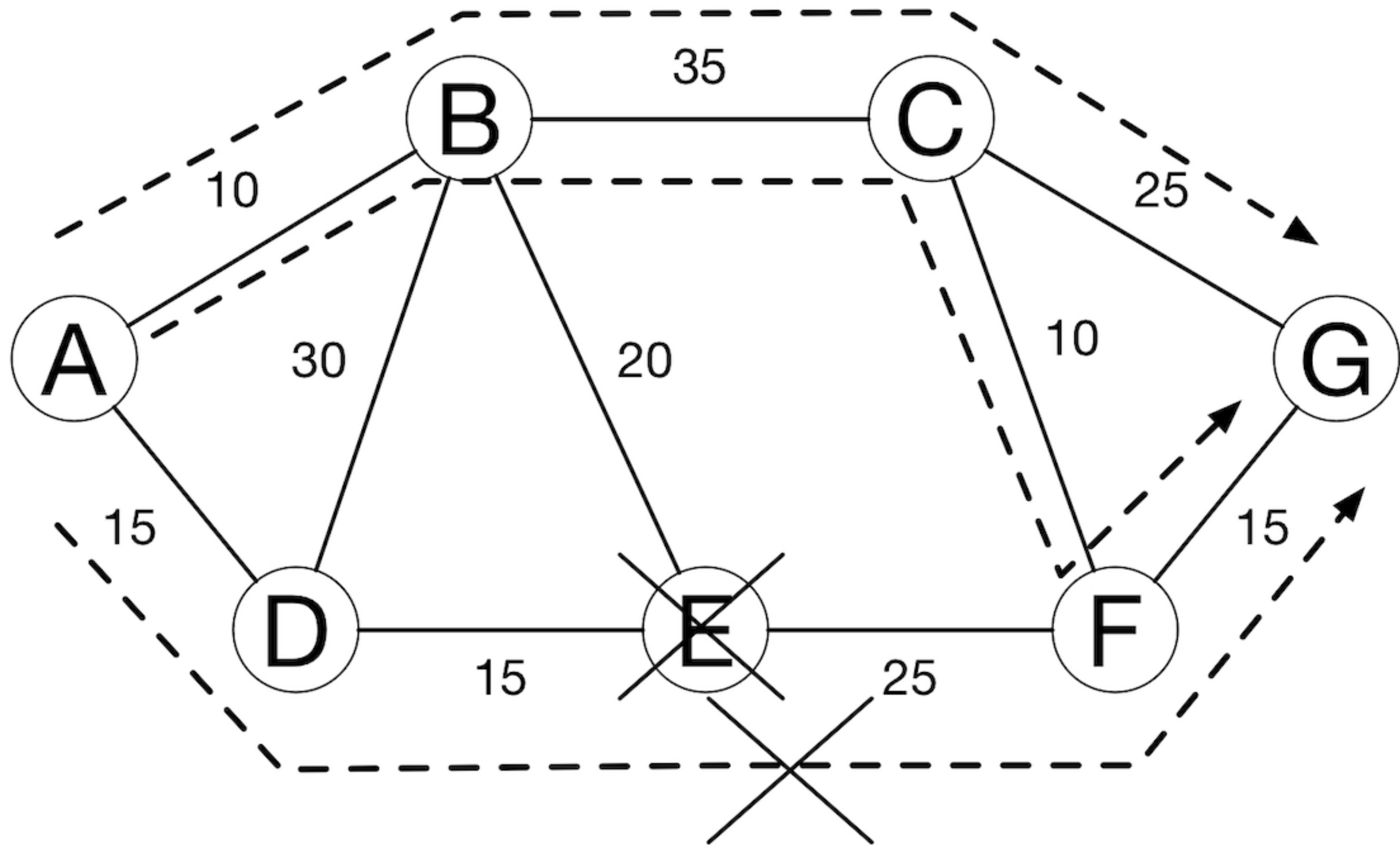
Simulating link failures

- **What if the link suddenly degrades or is disconnected?**
- **Largely increasing the cost of degraded or disconnected links will give an easy solution**



Administrative policies

- For many reasons, you don't want to accept packets from some nodes, depending on the relay paths
- For example: passing C is OK, but passing E is not: A-B-C-G and A-B-C-F-G are OK, but A-D-E-F-G is blocked
- Common among interconnection of the autonomous systems (internet service providers and organizations)



Routing information dissemination protocols

- **Link-state protocol: flooding link cost information of each node throughout the network**
- **Path vector protocol: exchanging path of nodes for each network instead of the link costs**
- **Highly vulnerable to external attacks**

Routing aggregation

- **The following four networks**
 - **192.168.100.0/24**
 - **192.168.101.0/24**
 - **192.168.102.0/24**
 - **192.168.103.0/24**
- **-> aggregated as 192.168.100.0/22**
- **4 networks together as one aggregated network**

Limitation of IP address structure

Network transports

IP address and the port number

- Each service has a 16-bit port number
- HTTPS = 443, DNS = 53, SSH = 22, etc.
- A pair of IP address and port number defines an endpoint of communication

UDP and TCP

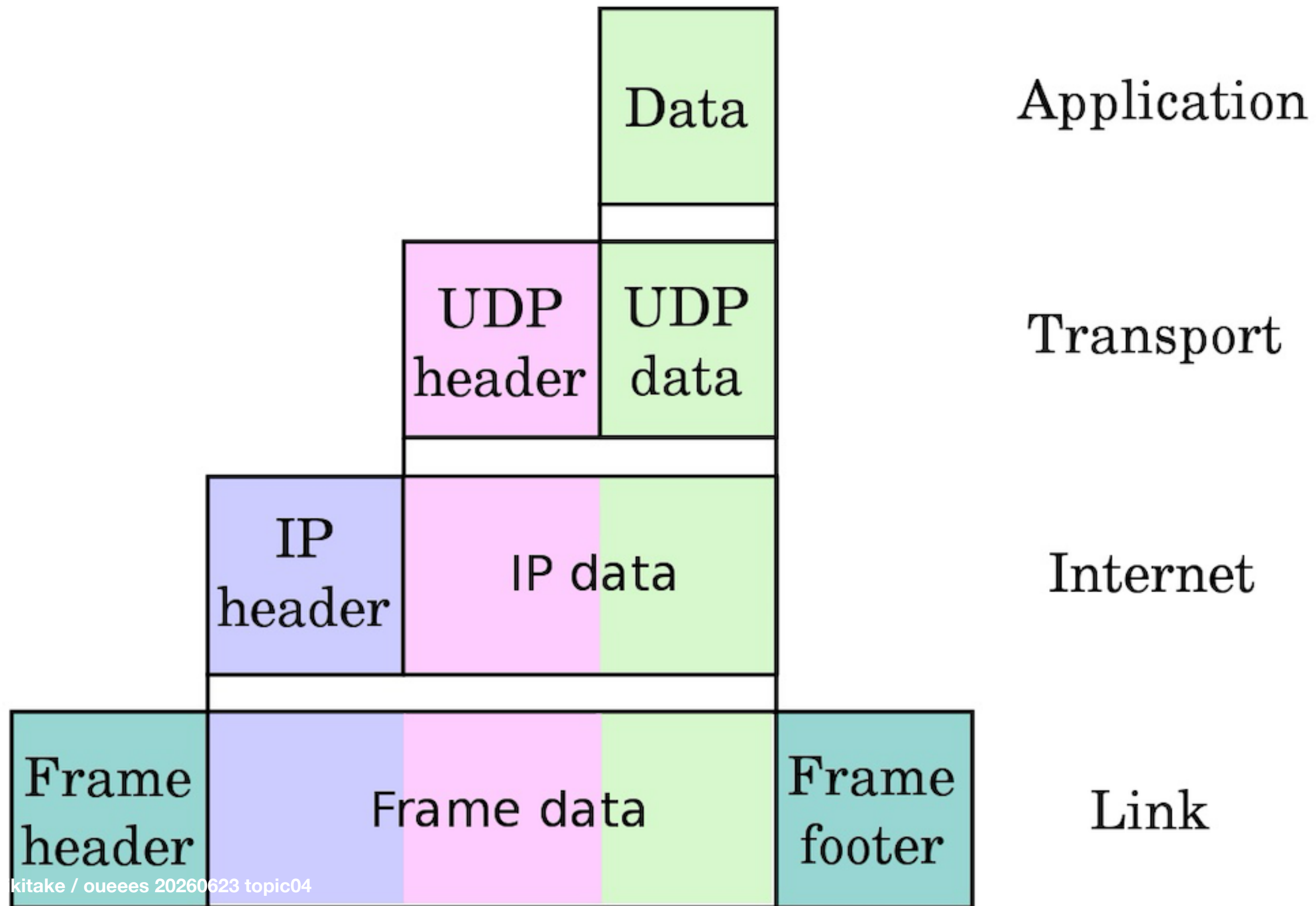
- Two major transport protocols on the internet
- User Datagram Protocol (UDP, RFC 768): connection-less
- Transmission Control Protocol (TCP, RFC 9293):
connection-oriented
 - Obsoleted RFCs: 793, 879, 2873, 6093, 6429, 6528, 6691
- See <https://www.rfc-editor.org> for all the internet RFCs
- RFC: Request for Comment

Packet exchange limitation

- **Packets are not always delivered**
- **Sending sequence is not preserved**
- **The same packet may be received multiple times**
- **The content of the packet may get altered or damaged**
- **Packet size has the limitation**

What UDP does

- Add a header with the port number
- Send it in an IP packet
- ... and that's it



UDP's pros and cons

- **UDP datagrams are still not always delivered and may get lost**
- **Sequence is not preserved**
- **The same datagram may be received multiple times and may cause duplicate delivery**
- **The errors in the contents of UDP datagrams are detectable**
- **UDP datagram has the size limit: suitable for relatively small messages**
- **Very small additional latency**

Transmission control protocol (TCP)

- Detect packet loss by timeout
- Split stream into segments
- Put sequence numbers to the segments
- Reassemble segments to the stream
- Perform congestion control



接收方

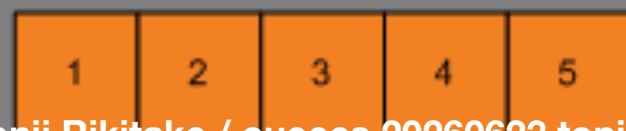


发送方

缓冲区



报文段为1460byte



序列号=1,数据: 1460 byte
序列号=1,确认号=1461,数据: 0 byte

序列号=1461,数据: 1460 byte

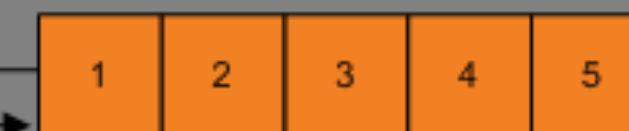
 序列号=2921,数据: 1460byte

序列号=4381,数据: 1460 byte

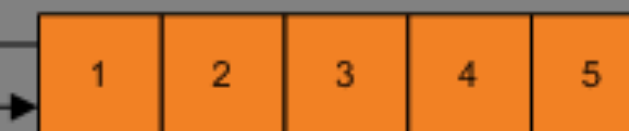
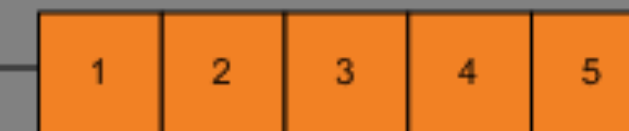
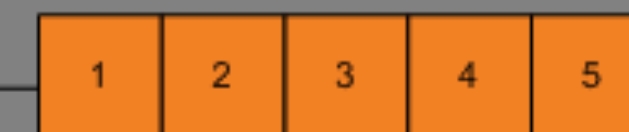
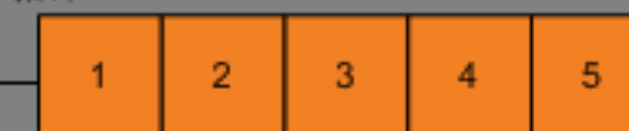
序列号=5841,数据: 1460 byte
序列号=1,确认号=2921,数据: 0 byte

序列号=2921,数据: 1460 byte
序列号=1,确认号=7301,数据: 0 byte

缓冲区



指针



时钟Timeout, 第三数据段被重发

TCP's pros and cons

- Loss is detected and recovered so long as the connection is alive
- Sequence is preserved
- No content repetition
- Errors are detected and fixed by retransmission
- The stream will accept data so long as the connection is alive
- Data delivery may delay if retransmission occurs

Web: HTTP/3: HTTP/2 over QUIC

- People wants *speed* and *smaller latency*
- HTTP/2 (RFC 7540): TCP-bound, stream aggregation and content compression
- QUIC (RFC 9000): UDP-based, tightly integrated to HTTP/2 and specific congestion control
- HTTP/2 had *head-of-line blocking problem* by TCP
- HTTP/3 (RFC 9114): HTTP/2 over QUIC (supported by most browsers already)

Buffering and head-of-line (HOL) blocking on HTTP/2¹

- Buffering causes only the oldest packets to be forwarded
- HTTP/2 uses single stream TCP for multiple HTTP requests and responses
- If TCP has retransmission, all the request-response exchanges will be halted until the lost data is recovered
- QUIC solved this problem by multiplexing multiple HTTP exchanges over the UDP datagrams

¹ Wikipedia contributors, [Head-of-line blocking: In reliable byte streams](#), Wikipedia, The Free Encyclopedia, 31 March 2024, 05:41 UTC [accessed 9 June 2024]

Transport for various network environments

- Local/Ethernet: short delay, low loss
- Mobile/WiFi: medium delay, high loss
- Longhaul/fiber: long delay, medium loss
- LEOs (Starlink): varying SNR, varying loss ²
- Interplanetary: ultra-long delay, ultra-high loss:
Delay Tolerant Network (DTN) ³

² Geoff Huston, [A Transport Protocol's View of Starlink](#), ISP Column, May 2024

³ Joab Jackson, [The Interplanetary Internet](#), IEEE Spectrum, 01 Aug 2005

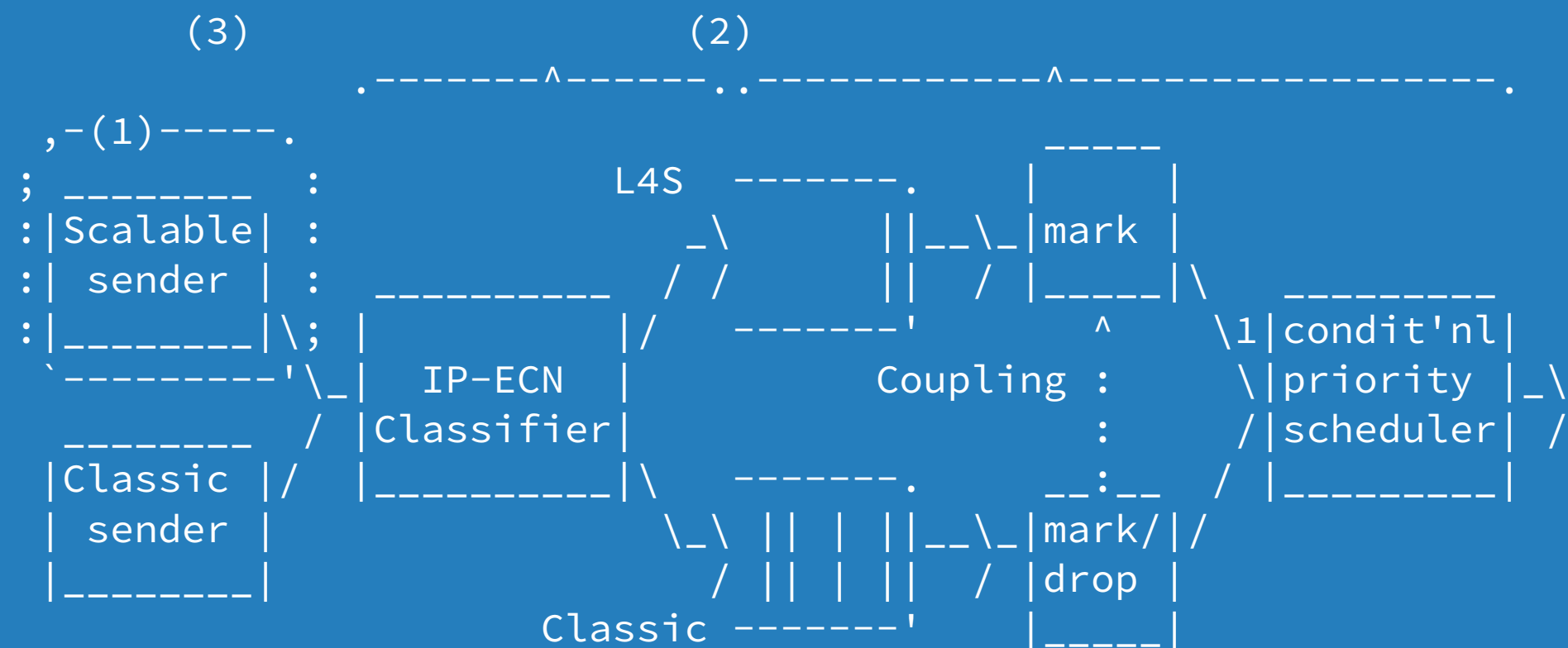
L4S for minimizing latency

- Demand for shorter latency/delay (1~2 ms)
- Major obstacle: queueing delays
- Data Center TCP (DCTCP)⁴
 - Key idea: use Explicit Congestion Notification for precise estimation of per-packet congestion
 - *Not* coexisting well with existing TCP
- L4S: Low Latency, Low Loss, and Scalable Throughput (L4S) Internet Service⁵
 - Coexistence of DCTCP and existing TCP in the same wide-area network

⁴ Stephen Bensley and Dave Thaler and Praveen Balasubramanian and Lars Eggert and Glenn Judd, **Data Center TCP (DCTCP): TCP Congestion Control for Data Centers**, RFC8257, October 2017

⁵ Bob Briscoe and Koen De Schepper and Marcelo Bagnulo and Greg White, **Low Latency, Low Loss, and Scalable Throughput (L4S) Internet Service: Architecture**, RFC9330, January 2023

L4S and the Active Queue Management (AQM) solution



- (1) Scalable sending host
- (2) Isolation in separate network queues
- (3) Packet identification protocol

Figure 1: Components of an L4S DualQ Coupled AQM Solution
(Quoted from RFC9330 Section 4.2 (Network Components))

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